
Chapter 12

Network Management

Outline

12.1 - Introduction

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12.3 - Configuration Management

12.4 - Performance and Fault Management

12.5 - End User Support

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12.1 Introduction

- **Network management**
 - Process of operating, monitoring and controlling the network to ensure that
 - Network working properly (efficient data exchange)
 - Providing value to its users
- **A lack of planning and organization**
 - Managers spend their time *firefighting* (reactive mode)
 - Dealing with breakdowns and immediate problems
- **Main areas of network management function:**
 - Configuration management
 - Performance and fault management
 - End user support
 - Cost management

Shift to LANs and the Internet

- **Large scale move from using mainframes and terminals to PCs, LANs and the Internet.**
 - Future of network management lies in the successful management of multiple clients and servers over LANs, BN's, and Internet
- **Focus on integration of organizational networks and applications. Main problems:**
 - Not all LANs use the same architecture
 - More types of network technology used, the more complex network management becomes
 - Staff must be familiar with many different types of devices

Integrating LANs, WANs and Internet

- **Keys integrating them into one organization**
 - Both LAN/Web and WAN managers to recognize that they no longer have total power
 - No more independent decision without considering the impact on other parts of network
 - Single overall goal that best meets the needs of the entire organizations
 - Require some managers to compromise on policies that may not be in the best interest of their own network
 - Must adopt a written charter to
 - Define its purpose, operational philosophy, and long range goals
 - Must develop individual procedures to implement policies

Integrating Voice & Data

- **Years ago, traditional voice and data networks (e.g., POTS and LANs) were handled by separate managers**
 - **Voice Communication Manager in Facilities Department:**
 - **Supervised telephone switchboard, coordinated installation and maintenance of the voice network**
 - **Data Communication Manager (IT department):**
 - **Installed own data circuit, installed and maintained computers**
- **Now, organizations realize benefits of integrating voice and data management function**
 - **Simplifies the network, and can lower network costs**
 - **Eliminates one department**
 - **Is now more typically found in network management**

12.3 Configuration Management

- **Major functions:**
 - **Management of the network's hardware and software configuration**
 - **Documenting the configuration**
 - **Ensuring that documentation is updated when there is a change in the configuration**

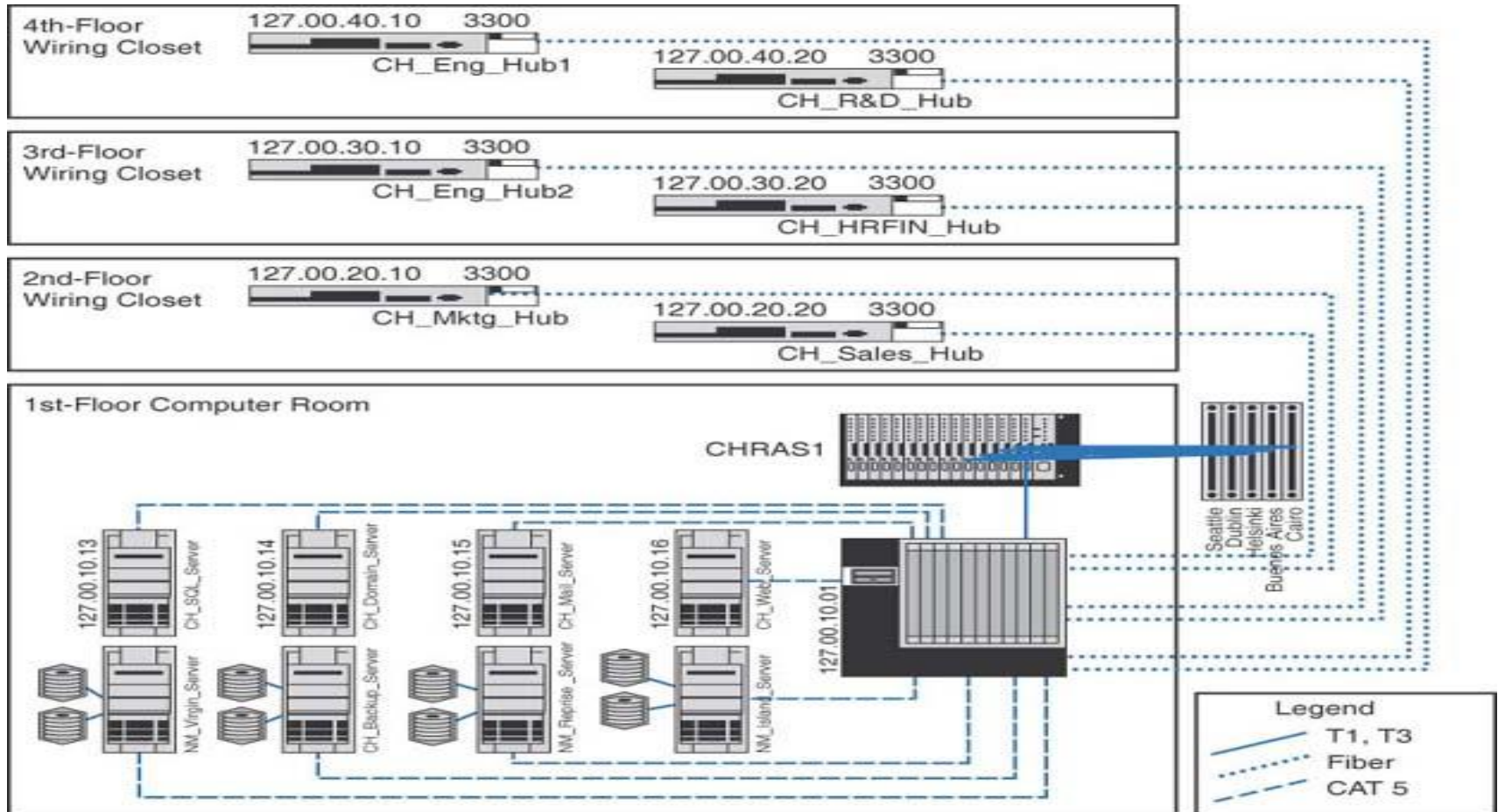
Configuring Network and Clients

- **Adding and deleting user accounts**
 - **New user addition**
 - **Categorized as a member of a group**
 - **With corresponding privileges**
 - **Given a standard login script**
 - **Commands to run when user logs in**
- **Updating software on client computers**
 - **Usually done automatically by Electronic Software Distribution (ESD) over the network at predefined time**
 - **Reduces cost of configuration management (long run)**
 - **Eliminates manual updates for each client**
 - **Produces accurate documents and reports**
 - **Being standardized: Desktop Management Interface (DMI)**

Documenting Configuration

- Includes information on
 - Network hardware and software
 - User and application profiles
 - Network documentation
- Documenting network hardware
 - Typically done via a set of diagrams documenting
 - Circuits, servers, devices, client computers
 - Must be supplemented with details on each device
 - Serial number, vendor, date of purchase, warranty information, repair history, phone number for repairs, etc
 - Any other info (e.g., contact info of managers)

Network Configuration Diagram



Documenting Configuration (Cont.)

- **Documenting software includes information on**
 - Network OS and any special-purpose network software
 - Version, release date, etc
 - Can be helpful in negotiating site licenses
- **Documentation of User & Application profiles**
 - Used to identify access rights to files and directories
- **Other network documentation**
 - Manuals for hardware, software, standards and operations; vendor contracts, and licenses
- **Documentation should include details about**
 - Performance and fault management, maintenance guidelines, DRP, user support and cost management

12.4 Performance & Fault Mgmt

- **Performance management**
 - Ensuring the network is operating as efficiently as possible
- **Fault management**
 - Preventing, detecting, and correcting faults in the network circuits, hardware, and software
- **Both require**
 - Network monitoring
 - Tracking the operation of network circuits and devices
 - Determining how heavily circuits and devices are being used
 - Ensuring circuits and devices are operating properly

Network Monitoring

- **Done by network management software**
- **Involves Network Operations Centers (NOCs)**
 - A centralized place enabling managers to monitor the network and respond to problems before users complain
- **Collect operational statistics from devices**
 - Physical network statistics
 - Derived from modems, multiplexers, and circuits linking hardware devices
 - Logical network information:
 - Derived from performance measurement systems
 - User response times, traffic volume on a specific circuit, the destinations of network packets, and other indices showing the network's service level

Failure Control Function

- **Requires developing a central control process for problem reporting and troubleshooting**
 - Maintain a central phone number for users to report problems
 - Establish a central troubleshooting group
 - Functions include contacting hardware, software vendors or common carriers
- **Detecting and reporting network problems**
 - By the help desk or NOC; by the user; and
 - By managed devices
 - Record data on messages processed and send this info back to a central database
 - Detect its own faulty parts and send alarms

Trouble Tickets

- **Reports that indicate network faults**
 - Automatically produced by software packages
- **Main reasons for trouble tickets**
 - Problem tracking
 - To determine status for correcting problems
 - Responsible people, schedule, status, resolution, etc
 - Problem statistics
 - Used to generate various statistical reports (number of failures per card, circuit, hardware, etc.,)
 - Problem prioritizing
 - To ensure critical problems get higher priority
 - Management reports
 - To determine availability, product and vendor reliability, vendor responsiveness

Trouble Log

- **Purpose**
 - To record problems that must be corrected and keep track of associated statistics
- **Helpful for reviewing problem patterns on the network**
- **Can be used to identify which network components are the most problematic**
- **Example of a trouble log:**
 - 37 calls for software problems, 26 calls for modems, 49 calls for client computers, 2 calls for common carriers

Performance and Failure Statistics

- **Main performance statistics**
 - Number of packets moved on a circuit and
 - Response time
- **Failure statistics**
 - **Availability:**
 - **Percent of time the network available to users**
 - Number of hours per month the network is available divided by the total of hours in a month (i.e., 720 hours)
 - Most networks: 99% to 99.5%
 - **Downtime:**
 - **Percent of time network is not available**

Failure Statistics

- **Mean time between failures (MTBF)**
 - Number of hours or days of continuous operation
 - Indicates the reliability of a component
- **Mean time to repair (MTTR)**
 - Average number of minutes or hours until the failed device is operational again

$$\text{MTTR}_{\text{Repair}} = \text{MTTDiagnose} + \text{MTTRespond} + \text{MTTFix}$$

↑
Average number of minutes until the root cause is diagnosed (shows efficiency of NOC)

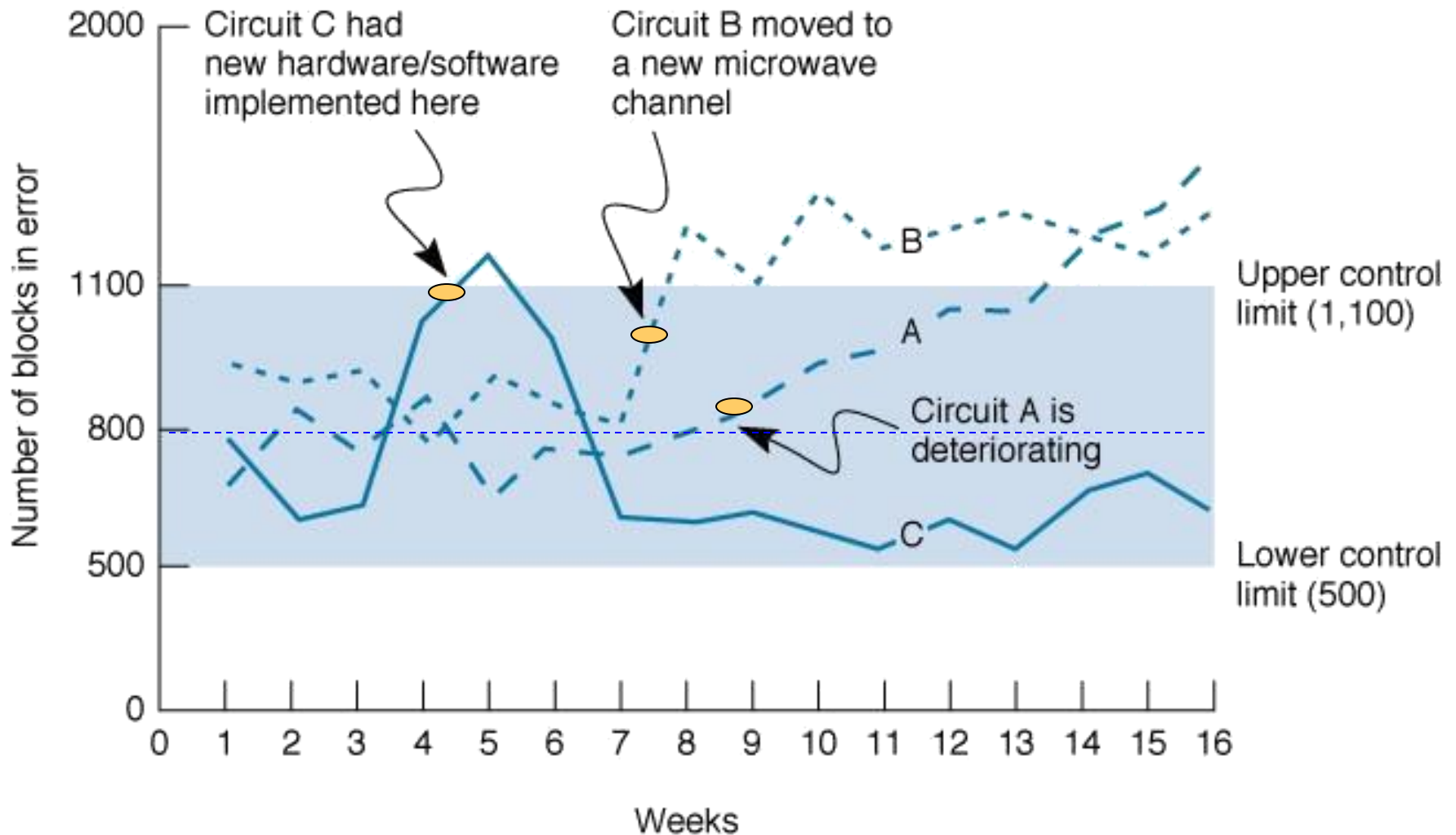
↑
Average number of minutes until the service or vendor personnel arrives at location

↑
Average number of minutes until problem is fixed (shows efficiency of repair people)

Other Daily Statistics

- **Daily data about the normal operations**
 - Number of errors per circuit
 - Volume of transmission per circuit, computer, etc.,
 - Usage rates; % of capacity used
- **Used to identify devices or circuits that have higher than average rates (thus may be good candidate for upgrades)**
- **Used for predicting future growth patterns and failures**
 - May be accomplished by establishing quality control charts that track network performance against an upper and a lower limits of a statistics

Quality Control Chart



Improving Performance

- **General activities to improve performance that cut across the different types of networks:**
 - **Policy-based management**
 - **Server load balancing**
 - **Service-level agreements**

Policy-Based Management

- **A new approach in managing performance**
- **Setting priority policies for network traffic in software**
 - **Take effect when the network becomes busy**
- **Example**
 - **Manager: Sets videoconferencing traffic as the highest priority**
 - **Since delays will have the highest impact on the performance of that application**
 - **Policy management software: Configures devices using QoS capabilities in TCP/IP and/or ATM**
 - **So that videoconferencing traffic gets the highest priority when device becomes busy**

Service Level Agreements (SLAs)

- Signed between the organization and its service providers (ISP or common carriers)
- Specify the exact type of performance and fault conditions that the organization will accept
- Also specify compensations the service provider must provide if it fails to meet the SLA
- Examples of SLA statements:
 - Availability must be 99% or higher
 - MTBF for T1 circuits must be 120 days or more
 - Maximum allocable response time must be lower than 2 minutes

12.5 End User Support

- **Solving the problems users encounter while using the network**
- **Main functions within end user support:**
 - **Resolving problems**
 - **End user Training**

Resolving Problems

- **Major sources of problems with user equipment**
 - Hardware device failures, generally easiest to fix
 - Lack of user knowledge on proper operation, also easier to fix
 - Solved by discussing the problem with user and taking the person through the process step by step
 - Problems with software, software settings or software incompatibility, generally hardest to fix
 - Could be result of a bug or software not functioning properly
 - May requires expertise with that specific software
 - May require software upgrades

Process of Resolving Problems

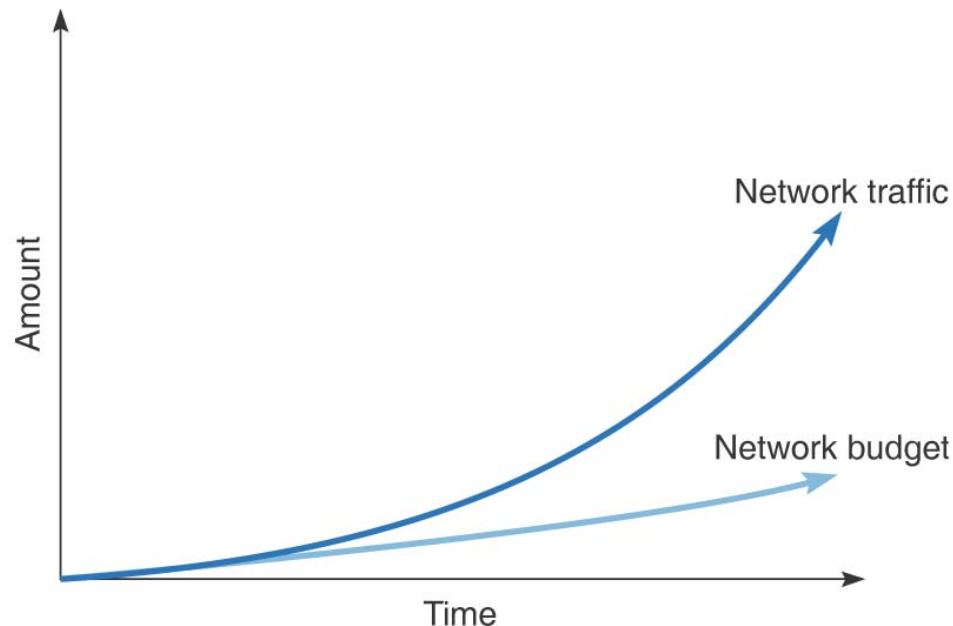
- Assistance requested by user from the help desk
- Trouble ticket created and maintained by the trouble log
- Problem resolved by the receiving staff members
 - If not resolved, problem is escalated
- Levels of problem resolution process
 1. Help desk: Resolve about 85% of problems
 2. Staff members with specialized skills specific to the problem at hand: Brought in when help desk can't solve
 3. Technical specialists with a higher level of training: Contacted by level 2 to look into harder problems

Providing End User Training

- **An ongoing responsibility of network manager**
- **A key part in implementation of new networks**
- **Important to have ongoing training programs**
 - **Employees often change jobs within an organization and new employees require training**
- **Types of training**
 - **In-class**
 - **Focus on in-depth instructions for the most commonly used functions in the network**
 - **One-on-one instructions**
 - **Through documentation and training manuals provided**

12.6 Cost Management

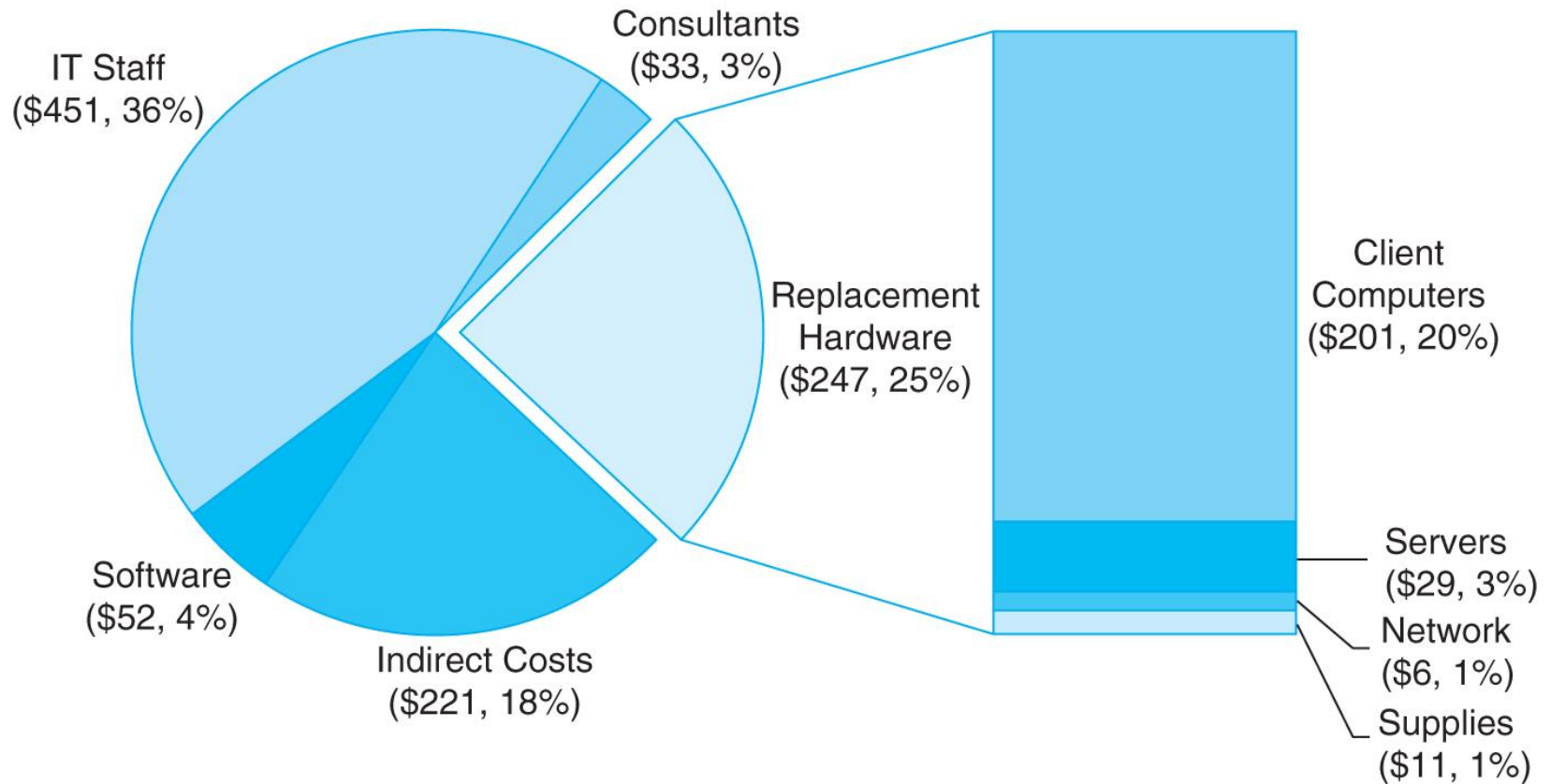
- One of the most challenging areas over the past few years
 - Traffic growing more rapidly than the budget
 - Managers are forced to provide greater capacity at an ever lower cost per megabyte



Sources of Cost

- **Total Cost of Ownership (TCO)**
 - A measure of how much it costs per year to keep one computer operating
 - Includes cost of
 - Repairs and software/hardware upgrades
 - Support staff (maintain, install, administer, etc)
 - Training and technical support
 - Time “wasted” by the user when problems occur
 - TCO of a Windows computer
 - TOO HIGH - \$5,000 and \$10,000 per computer per year
 - Main culprit in cost is inclusion of “wasted” time
 - Not commonly accepted method of accounting

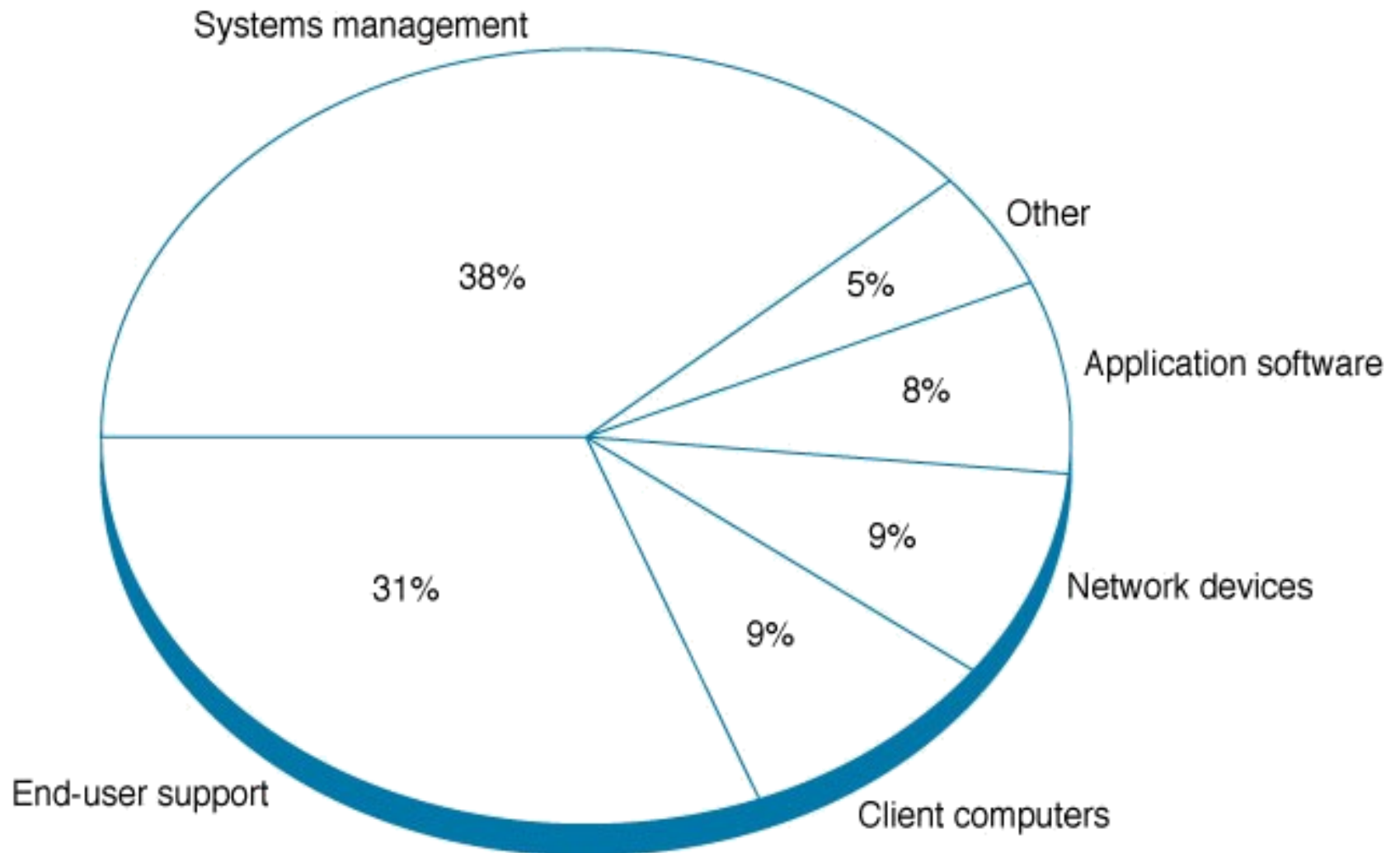
TCO per Client Computer for a School District



Sources of Cost (Cont.)

- **Net Cost of Ownership (NCO)**
 - An alternative to TCO; also called as “real TCO”
 - Measures only direct costs
 - Leaving out so-called “wasted” time
 - NCO costs per computer: between \$1500-\$3500
 - Net management for a 100-user network require an annual budget of between \$150,000 - \$350,000
 - Largest network budget items (with NCO):
 - Personnel cost, accounting for 50-70% of costs
 - WAN circuits
 - Hardware upgrades and replacement parts
 - Important message: Focus should be on ways to reduce personnel time, not hardware costs

Network Management Personnel Costs



Managing Network Budgets

- **Difficult to manage**
 - Networks growing rapidly → requiring more staff and more equipment
- **Use charge-back policies to counter increasing cost**
 - Allocate costs associated with the network to specific users
 - Users must pay for their usage (by moving part of their budget) to the network group
 - Applicable to WAN and mainframe users as a cost accounting mechanism
 - Difficult to implement on LANs

Reducing Costs

- **Develop standard hardware/software configurations for client computers, servers, and network devices**
 - Easier to diagnose and fix problems; Easier to learn
- **Automate as much of the network management process as possible**
 - Use of ESD, DHCP and a other network management tools
- **Reduce the cost of installing new hardware/software**
 - Reduce the staff time spent in installing; Work with vendors
- **Centralize help desks**
 - More specialists in one location → faster problem resolution
- **Evaluate move to thin client architectures**
 - Reduces software and support costs

12.7 Implications for Management

- **Network management requires**
 - A good understanding of networking technologies
 - An ability to work with end users and management
 - An understanding of key elements driving network costs
- **Requires special skill to explain the business value of the networks to senior management**
 - Needed to justify increased cost of management
- **Network management increasing its complexity**
 - New, more intelligent technologies in managing networks
 - More reliable, but more expensive and requires new skills in designing, installing and managing
 - Difficulty in keeping the staff